

Local Coverage Determination (LCD)

Cardiac Computed Tomography (CCT) and Coronary Computed Tomography Angiography (CCTA)

L33947

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Contractor Information

Contractor Name	Contract Type	Contract Number	Jurisdiction	States
CGS Administrators, LLC	MAC - Part A	15101 - MAC A	J - 15	Kentucky
CGS Administrators, LLC	MAC - Part B	15102 - MAC B	J - 15	Kentucky
CGS Administrators, LLC	MAC - Part A	15201 - MAC A	J - 15	Ohio
CGS Administrators, LLC	MAC - Part B	15202 - MAC B	J - 15	Ohio

LCD Information

Document Information

LCD ID
L33947

LCD Title

Cardiac Computed Tomography (CCT) and Coronary Computed Tomography Angiography (CCTA)

Proposed LCD in Comment Period

N/A

Source Proposed LCD

N/A

Original Effective Date

For services performed on or after 10/01/2015

Revision Effective Date

For services performed on or after 10/09/2025

Revision Ending Date

N/A

Retirement Date

N/A

Notice Period Start Date

N/A

Notice Period End Date

N/A

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Issue

Issue Description

This LCD outlines limited coverage for this service with specific details under Coverage Indications, Limitations and/or Medical Necessity.

CMS National Coverage Policy

Language quoted from Center for Medicare and Medicaid Services (CMS), National Coverage Determinations (NCDs) and coverage provisions in interpretive manuals is italicized throughout the policy. NCDs and coverage provisions in interpretive manuals are not subject to the Local Coverage Determination (LCD) Review Process (42 CFR 405.860[b] and 42 CFR 426 [Subpart D]). In addition, an administrative law judge may not review an NCD. See Section 1869(f)(1)(A)(i) of the Social Security Act.

Unless otherwise specified, italicized text represents quotation from one or more of the following CMS sources:

Title XVIII of the Social Security Act (SSA):

Section 1862(a)(1)(A) excludes expenses incurred for items or services which are not reasonable and necessary for the diagnosis or treatment of illness or injury or to improve the functioning of a malformed body member.

Section 1833(e) prohibits Medicare payment for any claim which lacks the necessary information to process the claim.

Section 1862 (a)(7) excludes routine physical examinations.

Code of Federal Regulations:

42 CFR Section 410.32 indicates that diagnostic tests may only be ordered by the treating physician (or other treating practitioner acting within the scope of his or her license and Medicare requirements) who furnishes a consultation or treats a beneficiary for a specific medical problem and who uses the results in the management of the beneficiary's specific medical problem. Tests not ordered by the physician (or other qualified non-physician provider) who is treating the beneficiary are not reasonable and necessary (see Sec. 411.15(k)(1) of this chapter).

CMS Publications:

CMS Publication 100-3, *National Coverage Determination Manual*, Chapter 1
220.1 Computerized Tomography

CMS Publication 100-4, *Medicare Claims Processing Manual*, Chapter 13
20 Payment Conditions for Radiology Services

CMS Publication 100-9, *Contractor Beneficiary and Provider Communication Manual*,
Chapter 5
20 Correct Coding Initiative

Coverage Guidance

Coverage Indications, Limitations, and/or Medical Necessity

Abstract:

The multidetector helical computed tomography (MDCT) technology requires thin (up to 1 mm) slices, 0.5 to 0.75 mm reconstructions, multiple simultaneous images (e.g. 16, 32, 64 or more slices), and cardiac gating (often requiring beta blockers for ideal heart rate). There is significant post-processing, depending on the number of slices per second for image

generation. For coronary artery imaging, the resulting images show a high correlation with stenotic lesions noted on diagnostic cardiac catheterization but more importantly, with atheromas on intracoronary ultrasound.

Current available body of evidence demonstrates that CCTA can reliably rule out the presence of significant coronary artery disease (CAD) in patients with a low to intermediate probability of having CAD and can reliably achieve a high degree of diagnostic accuracy and technical performance necessary to replace conventional angiography.

Indications:

- a. Patient presenting with chest pain syndrome.

CCTA may be used in lieu of an imaging stress test. The clinician must have a high degree of suspicion that CAD is high on the differential diagnosis of the symptoms.

- b. To facilitate the management decision of a patient with an equivocal stress test.

CCTA might be chosen in select patients who have an equivocal stress (or stress imaging) test. The rationale is that a noninvasive coronary anatomic test (CCTA) allows an alternate method of assessing the coronary arteries, which would limit the number of negative invasive coronary angiograms.

- c. When the recurrence of symptoms in patients with known coronary artery disease may be related to progression/exacerbation of underlying disease.

The use of CCTA in this setting would be to evaluate the extent of previously diagnosed coronary artery disease. Patients with known disease may have had remote invasive angiography and/or stress testing to evaluate prior events or symptoms. New or recurrent symptoms may relate to a change in the coronary anatomy that can be assessed with CCTA.

- d. When patients with prior bypass surgery or intracoronary artery stent placement present with chest pain or dyspnea.

Coronary bypass grafts are relatively well seen with CCTA. The rationale for CCTA would be to determine the patency and severity of possible graft stenoses that may be the source of chest pain. Patients with prior intracoronary stents often present with recurrent chest pain. The rationale for a CCTA as an alternative to invasive angiography is to rule out in-stent restenosis as the cause of symptoms. (Accurate assessment of in-

stent restenosis may be limited by the artifact caused by the stent material itself and the quality of the scan and scanner).

e. Suspected congenital anomalies of the coronary circulation.

CCTA is used to assess patients suspected of having a congenital coronary anomaly. The cross-sectional nature of this technique allows one to determine accurately both the presence and possible future harm that could result from the anomaly. It is often used after an anomaly has been identified following a different test such as prior invasive coronary angiogram. A CCTA is used to decide if surgery is indicated and for surgical planning.

f. The assessment of coronary or pulmonary venous anatomy.

This application of CTA for the coronary and pulmonary veins is primarily for pre-surgical planning. Coronary venous anatomy can be useful for the cardiologist who needs to place a pacemaker lead in the lateral coronary vein in order to resynchronize cardiac contraction in patients with heart failure. This may be helpful to guide biventricular pacemaker placement.

Pulmonary vein anatomy can vary from patient to patient. Pulmonary vein catheter ablation can isolate electrical activity from the pulmonary veins and allow for the elimination of recurrent atrial fibrillation. The presence of a pulmonary venous anatomic map may help eliminate procedural complications and allow for the successful completion of the procedure.

g. The patient undergoing non-coronary artery cardiac surgery.

Certain patients who have non-coronary artery cardiac surgery (valve or ascending aortic surgery) may need a pre-operative invasive coronary angiogram. The surgical planning may also depend upon the exact location of the coronary arteries. The rationale for the use of CCTA in these patient subsets is to avoid potentially unnecessary invasive testing and still provide appropriate pre-surgical information.

h. The test may be medically necessary in patients presenting to the emergency room with complaints consistent with cardiac ischemia, but without diagnostic electrocardiography (ECG) or enzymes.

i. The test may be considered medically necessary in patients status post revascularization procedures who present with recurrent symptoms not clearly

identifiable as ischemic. **Limitations:**

- a. The test is never covered for screening, i.e., in the absence of signs, symptoms or disease.
- b. The test will be considered not medically necessary if the anticipated results are not expected to provide new, additional information to that already previously obtained from other tests (such as stress myocardial perfusion images or cardiac ultrasound). New or additional information should facilitate the management decision, not merely add a new layer of testing.
- c. For dates of service prior to 01/01/2010, determination of cardiac ejection fraction (CPT code 0151T) should not be billed when previously determined by other techniques. CPT code 0151T is deleted effective 12/31/2009.
- d. The test will be considered not medically necessary if it is anticipated that the patient would require invasive cardiac angiography for further diagnosis or for therapeutic intervention. (e.g., angina decubitus, unstable angina, Prinzmetal angina, etc.)
- e. The test may be denied, on post-pay review, as not medically necessary when used for cardiac evaluation if there were pre-test knowledge of sufficiently extensive calcification of the suspect coronary segment that would diminish the interpretive value.
- f. Effective 12/01/2009, coverage for evaluation of coronary artery or bypass graft stenosis, or for functional status (e.g., wall motion), is limited to multidetector scanners having at least 64 slices per rotation capability. This two year period (12/01/2007 - 12/01/2009) will allow for a phase-in of new technology.
- g. The administration of beta blockers and the monitoring of the patient during MDCT/CCTA by a physician experienced in the use of cardiovascular drugs is included as part of the test and is not a separately payable service.
- h. All studies must be ordered by the physician/qualified non-physician practitioner treating the patient and who will use the results of the test in the management of the patient.
- i. The test must be performed under the direct supervision of a physician.
- j. This LCD does not address electron beam tomography (EBT) technology or Ultrafast CT for coronary artery examination. There is no extension of coverage of EBT based on this policy.
- k. Quantitative calcium scoring (CPT code 0144T for dates of service prior to 01/01/2010, and CPT 75571 on or after 01/01/2010) is not a covered service and will be denied as not medically necessary. Calcium scoring reported in isolation is considered a screening service. When performed in association with CT angiography, there is neither separate nor additional included reimbursement for the calcium scoring.

I. Acceptable Levels of Competence for Performance and Interpretation: Providers submitting claims for these tests must demonstrate proficiency and training in performing the tests according to the following standards:

The acceptable levels of competence, as defined by the American College of Cardiology (ACC)/American Heart Association (AHA) Clinical Competence Statement on Cardiac Imaging with Computed Tomography and Magnetic Resonance (2005) and the American College of Radiology (ACR) Clinical Statement on Noninvasive Cardiac Imaging (2005), are outlined as follows:

For the technical portion

, a recommended level of competence is fulfilled when the image acquisition is obtained under all of the following conditions:

1. The service is performed by a radiology technologist who is credentialed by a nationally recognized credentialing body (American Registry of Radiologic Technologists or equivalent) and meets state licensure requirements where applicable.
2. If intravenous beta blockers or nitrates are to be given prior to a CT coronary angiogram, the test must be under the direct supervision of a certified registered nurse and physician (familiar with the administration of cardiac medications) who are available to respond to medical emergencies and it is strongly recommended that the certified register nurse and physician be ACLS certified.
3. When contrast studies are performed, the physician must provide direct supervision and the radiologic technologist or registered nurse administering the contrast must have appropriate training on the use and administration of contrast media.

For the professional portion

, a recommended level of competence is fulfilled when the interpretation is performed by a physician meeting one of the following requirements:

1. The physician has appropriate additional training in coronary CTA and cardiac CT imaging equivalent to the guidelines set forth by the ACC or ACR (for example: the ACCF/AHA Clinical Competence Statement on Cardiac Imaging with Computed Tomography and Magnetic Resonance (2005) and the ACR Clinical Statement on Noninvasive Cardiac Imaging (2005)), or

2. The physician has appropriate medical staff privileges to interpret CT coronary angiograms at a hospital that participates in the Medicare program, and is actively training in cardiac CT (as in paragraph a). A grace period of 24 months will be allowed to acquire the necessary training.

Summary of Evidence

N/A

Analysis of Evidence (Rationale for Determination)

N/A

General Information

Associated Information

N/A

Sources of Information

This bibliography presents those sources that were obtained during the development of this policy. CGS is not responsible for the continuing viability of Web site addresses listed below.

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Model Local Coverage Determination (LCD) Work Group for Cardiac Computed Tomography (CCT) and Computed Tomography Coronary Angiography (CTCA), comprising of the American College of Cardiology (ACC), Carrier Advisory Committee (CAC), American College of Radiology (ACR), American Society of Nuclear Cardiology (ASNC), North American Society for Cardiac Imaging (NASCI) Society of Cardiac Angiography and Intervention (SCAI) and Society of Cardiovascular CT (SCCT), which had the following sources:

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Bibliography

N/A

Revision History Information

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
10/09/2025	R18	Revision Effective: 10/09/2025	• Other (Annual Review)

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		Revision Explanation: Annual review, no changes.	
08/07/2025	R17	Revision Effective: 08/07/02025 Revision Explanation: Added 'one of 'to <i>a recommended level of competence is fulfilled when the interpretation is performed by a physician meeting the following requirements:'</i> under For the professional portion in the coverage and indications section.	• Provider Education/Guidance
10/03/2024	R16	R16 Revision Effective: 10/03/2024 Revision Explanation: Annual Review, no changes 09/27/2024: <i>At this time 21st Century</i>	• Other (Annual Review)

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		<i>Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	

10/05/2023

R15

R15

- Other (Annual Review)

Revision Effective:
10/05/2023

Revision
Explanation: Annual
Review, no changes

09/29/2023: *At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage*

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		<i>determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	
10/06/2022	R14	R14 Revision Effective: 10/06/2022 Revision Explanation: Annual Review, no changes 09/26/2022 :<i>At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	• Other (Annual Review)

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
09/30/2021	R13	R13 Revision Effective: 09/30/2021 Revision Explanation: Annual Review, no changes 09/22/2021 :<i>At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	Other (Annual Review)
09/19/2019	R12	R12 Revision Effective: N/A Revision Explanation: Annual Review, no changes	Other (Annual Review)

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		09/30/2020 :At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.	

09/19/2019

R11

Annual Review

- Other (R11

Revision Effective:
11/07/2019

Revision Explanation:
Annual Review, Removed other comments from Coverage Indications, Limitations and/or Medical Necessity and Associated Information based on TDL 190550. Added to A56451 - Billing and Coding: Cardiac Computed Tomography (CCT) and Coronary Computed Tomography

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
			Angiography (CCTA).
			10/30/2019:At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.)
09/19/2019	R10	R10 Revision Effective: 09/19/2019 Revision Explanation: Converted policy into new policy template that no longer includes coding section based on CR 10901. For Approval, no changes. 09/13/2019:At this time 21st Century Cures Act will apply to new and revised LCDs that restrict	Revisions Due To Code Removal

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		<i>coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	

09/19/2019

R9

R9

- Revisions Due To Code Removal

Revision Effective:
09/19/2019

Revision
Explanation:
Converted policy into new policy template that no longer includes coding section based on CR 10901.

09/12/2019: *At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the*

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
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coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.

04/04/2019	R8	R8 Revision Effective: 04/04/2019 Revision Explanation: Correct typographical error. 03/29/2019: <i>At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	• Typographical Error
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04/04/2019	R7	R7 Revision Effective: 04/04/2019 Revision Explanation: Removed all billing and coding details from	• Other (Removed billing and coding details)
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Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		policy into related Billing and Coding article. Coding information was removed based on CR10901. 03/29/2019: <i>At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	
10/01/2016	R6	Revision #: R6 Revision Effective: N/A Revision Explanation: annual review no changes made. <i>10/30/2018: At this time 21st Century Cures Act will apply to new and revised</i>	Other (Annual Review)

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		<i>LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the fields included on the LCD are applicable as noted in this policy.</i>	

10/01/2016

R5

Revision #: R5

Revision Effective:

N/A

Revision

Explanation: annual review no changes made.

DATE

(10/30/2017): At this time 21st Century Cures Act will apply to new and revised LCDs that restrict coverage which requires comment and notice. This revision is not a restriction to the coverage determination; and, therefore not all the

- Other (Annual Review)

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
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fields included on the LCD are applicable as noted in this policy.

10/01/2016	R4	Revision #: R4 Revision Effective: N/A Revision Explanation: annual review no changes made.	Other (Annual Review)
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10/01/2016	R3	Revision #: R3 Revision Effective: 10/01/2016 Revision Explanation: Q25.2 and Q25.4 deleted ICD-10 annual update and replaced with the following codes: Q25.21, Q25.29, and Q25.41-Q25.49.	Revisions Due To ICD-10-CM Code Changes
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Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
10/01/2015	R2	Revision #: R2 Revision Effective: N/A Revision Explanation: annual review no changes made.	Other (Annual Review)
10/01/2015	R1	Revision #: R1 Revision Effective: N/A Revision Explanation: Accepted the revenue code description changes.	Other (revenue code description changes)

Associated Documents

Attachments

N/A

Related Local Coverage Documents

Articles

[A56451 - Billing and Coding: Cardiac Computed Tomography \(CCT\) and Coronary Computed Tomography Angiography \(CCTA\)](#) 

Related National Coverage Documents

NCDs

[220.1 - Computed Tomography](#) 

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